

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)		
Programme: B.Sc. Information Technology	Class: TY	Semester: V
Exam: CIA	Name of the Course: Big Data and NOSQL	
Date of Exam: 29/07/2025	Marks: 20	Time: 40 Minutes
Instructions: 1) Make suitable assumptions wherever necessary and state the assumptions made. 2) Draw neat labelled diagrams wherever necessary.		

Q Attempt any four (5 marks each)

1. Explain the three V's of Big Data —Volume, Variety, and Velocity — and describe how each of these factors influences data processing and analysis.
2. What is the CAP theorem? Provide a brief explanation of its three properties: Consistency, Availability, and Partition Tolerance.
3. Differentiate between SQL and NoSQL databases. Discuss the key differences in terms of structure, scalability, and use cases.
4. In MongoDB, explain the concepts of _id, capped collections, and BSON. How does each of these contribute to efficient data storage and retrieval?
5. Demonstrate how to explicitly create a collection in MongoDB. In addition, explain the usage of selectors and projectors in querying data, and provide an example.
6. What is the purpose of the findOne() method in MongoDB? Also, explain how the explain() function can be used to optimize query performance.

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